



PARALA MAHARAJA ENGINEERING COLLEGE

(An Autonomous college affiliated to Biju Patnaik University of Technology, Odisha, Rourkela)

SITALAPALLI, BERHAMPUR, DIST.: GANJAM, PIN-761003

CSE PRODUCT DEVELOPMENT LAB

6TH SEMESTER

ACTIVITIES TO BE DONE BY STUDENTS IN DESIGN PHASE

S. No.	Design Activity	What Students Should Do	Key Deliverables
1	System Architecture Design	Select architecture style (MVC, Client–Server, Microservices, etc.), define system components and interactions	High-level architecture diagram, tech stack justification
2	Module Decomposition	Break system into logical modules with defined responsibilities and interfaces	Module description document
3	UML Modeling	Model system behavior and structure using standard UML diagrams	Use Case, Class, Sequence, Activity diagrams
4	Database Design	Identify entities, relationships, keys, constraints; normalize data	ER diagram, database schema
5	Algorithm & Logic Design	Develop pseudocode, flowcharts, choose data structures and algorithms	Pseudocode, flowcharts, complexity reasoning
6	UI/UX Design	Design wireframes, screen flow, and user interaction structure	Wireframes, navigation flow diagram
7	Technology Stack Finalization	Select and justify programming language, frameworks, tools, and deployment platform	Technology justification document
8	Non-Functional Design	Define performance, security, scalability, reliability, and error handling strategies	NFR specification section
9	Test Design (Pre-coding)	Identify test cases, boundary conditions, input–output validation scenarios	Test case table

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